

Součtové a rozdílové vzorce

$$\begin{aligned}\sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta \\ \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \\ \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}\end{aligned}$$

Součiny $\rightarrow$ součty

$$\begin{aligned}\sin \alpha \sin \beta &= \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)] \\ \sin \alpha \cos \beta &= \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)] \\ \cos \alpha \cos \beta &= \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)]\end{aligned}$$

Součty $\rightarrow$ součiny

$$\begin{aligned}\sin \alpha + \sin \beta &= 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} \\ \cos \alpha + \cos \beta &= 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} \\ \cos \alpha - \cos \beta &= -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}\end{aligned}$$

Dvojnásobný úhel

$$\begin{aligned}\sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= 1 - 2 \sin^2 \alpha = 2 \cos^2 \alpha - 1 \\ \tan 2\alpha &= \frac{2 \tan \alpha}{1 - \tan^2 \alpha}\end{aligned}$$

Poloviční úhel

$$\begin{aligned}\sin^2 \frac{\alpha}{2} &= \frac{1 - \cos \alpha}{2}, \quad \cos^2 \frac{\alpha}{2} = \frac{1 + \cos \alpha}{2} \\ \tan \frac{\alpha}{2} &= \frac{\sin \alpha}{1 + \cos \alpha} = \frac{1 - \cos \alpha}{\sin \alpha}\end{aligned}$$

Základní identity

$$\sin^2 \alpha + \cos^2 \alpha = 1, \quad 1 + \tan^2 \alpha = \frac{1}{\cos^2 \alpha}, \quad 1 + \cot^2 \alpha = \frac{1}{\sin^2 \alpha}$$

Vyjádření přes $\tan$

$$\sin \alpha = \frac{\tan \alpha}{\sqrt{1 + \tan^2 \alpha}}, \quad \cos \alpha = \frac{1}{\sqrt{1 + \tan^2 \alpha}}$$

Lineární kombinace

$$\begin{aligned}\sin x + \cos x &= \sqrt{2} \cos\left(\frac{\pi}{4} - x\right) \\ \sin x - \cos x &= \sqrt{2} \sin\left(x - \frac{\pi}{4}\right)\end{aligned}$$

Základní goniometrické rovnice

$$\begin{aligned}\sin x = 0 &\Rightarrow x = k\pi \\ \cos x = 0 &\Rightarrow x = \frac{\pi}{2} + k\pi \\ \tan x = 0 &\Rightarrow x = k\pi \\ \sin x = 1 &\Rightarrow x = \frac{\pi}{2} + 2k\pi \quad (k \in \mathbb{Z}) \\ \sin x = -1 &\Rightarrow x = -\frac{\pi}{2} + 2k\pi \\ \cos x = 1 &\Rightarrow x = 2k\pi \\ \cos x = -1 &\Rightarrow x = \pi + 2k\pi\end{aligned}$$